Exam - Data Analysis

FIRS - AgroParisTech - Géodata Paris

Data Analysis and Storytelling - Evaluation

Data Analysis Part

Last Name:

First Name:

Cross the correct answer for the 10 following questions (1 point per question).

Question 1

In the following equation for an univariate linear model:

$$y_i = \beta_0 + \beta_1 * x_i + \epsilon_i, \epsilon_i \sim N(0, \sigma)$$

What is the ϵ term?

- [] The linearity coefficient
- [] The uncertainty around the regression coefficient
- [] The difference between the observation and the regression line
- [] The average y values

Question 2

We fitted a model similar to the equation presented in Q1 and got the following output:

```
Call:
lm(formula = y ~ x)

Residuals:
     Min       1Q   Median       3Q      Max
-2.23797 -0.61323 -0.01973  0.59633  2.21723

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.94612    0.09693   51.03   <2e-16 ***
x            0.97754    0.08546   11.44   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9693 on 98 degrees of freedom
Multiple R-squared:  0.5718,    Adjusted R-squared:  0.5674
F-statistic: 130.9 on 1 and 98 DF,  p-value: < 2.2e-16
```

What are the estimated values for the parameters β_0, β_1, σ (rounded to the 2nd decimal)?

- [] 4.95, 0.98, 0.97
- [] 0.97, 4.95, 0.57
- [] 130.9, 4.95, 0.98
- [] 4.95, 0.98, 0.57

Question 3

Assume that we fitted an univariate model and got the following equation:

$$y = 3 - 0.8 * x + \epsilon, \epsilon \sim N(0, 0.9)$$

What is the correct interpretation for the slope parameter - β_1 ?

- [] For every change in one unit of x, y increase by 0.8
- [] For every change in one unit of x, y decrease by 3
- [] For every change in one unit of x, y decrease by 0.8
- [] For every change in one unit of x, y increase by 3

Question 4

The following two models were fitted:

```
m1 <- lm(y ~ x, data = dat)
m2 <- lm(y ~ x + z, data = dat)
```

The estimated regression coefficients are for m1:

```
coef(m1)
```

```
(Intercept)           x
  4.8898645  -0.5067707
```

and for m2:

```
coef(m2)
```

```
(Intercept)           x           z
   4.946508    1.018086   -2.007411
```

Why is the estimated coefficient for x different - the sign change - between the two models?

- [] x and z are likely correlated, in m1 the estimated coefficient for x absorb part of the negative effect of z
- [] x and z are likely uncorrelated, in m1 the variance component for x is inflated
- [] due to type I error rate, x effect is underestimated in m1 compared to m2
- []

Question 5

We have the following data.frame in R:

```
head(dat)
```

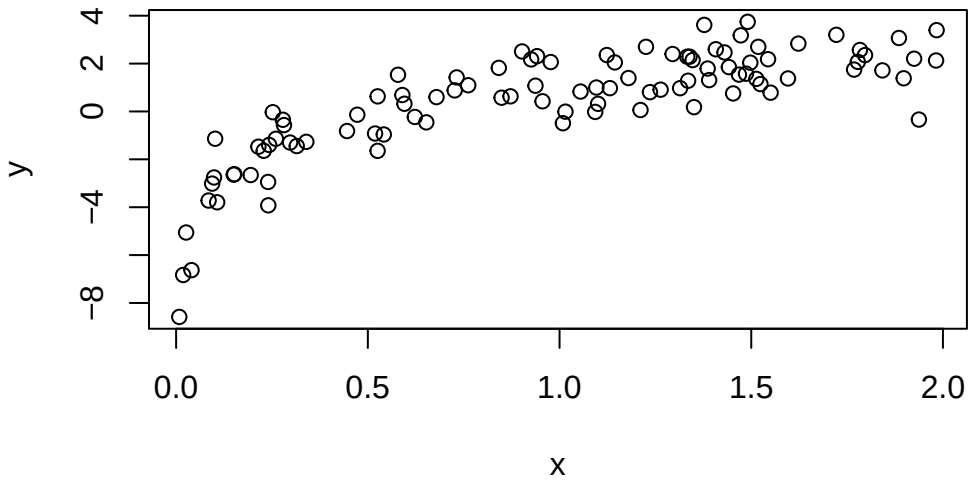
```
           x f          y
1  1.9442172 b -0.7152422
2 -1.4517301 d -0.7526890
3  1.6212383 d -0.9385387
4  0.3052074 b -1.0525133
5 -0.4182046 c -0.4371595
6 -0.2007901 c  0.3311792
```

What is the correct R synthax to fit a linear model with y as the response variable, x and f as covariable with an interaction term between the two?

- [] `m <- lm(y ~ x + f + x@f, data = dat)`
- [] `m <- lm(y ~ x + f, data = dat)`
- [] `m <- lm(y ~ x + f + x:f, data = dat)`
- [] `m <- lm(y ~ x * f, data = dat)`

Question 6

We observe the following relationship between a variable x and a variable y:



Which model call in R would be adapted to fit this relation?

- [] `lm(y ~ x + I(x^2))`
- [] `lm(y ~ x - 2)`
- [] `lm(y ~ log(x))`
- [] `lm(y ~ x)`

Question 7

Forest biodiversity is recorded across France in the plots of the French National Forest Inventory. We want to use a hierarchical model with a random intercept for the department to estimate

the average biodiversity at the level of the department. What is the correct equation for such a model (i is an index for the observation and j an index for the department)?

- [] $biodiversity_i = \beta_0 + \beta_1 * department_i + \epsilon_i$
- [] $biodiversity_i = \beta_0 + department_i + \epsilon_i$
- [] $biodiversity_{ij} = \beta_0 + \delta_j + \epsilon_i, \delta_j \sim N(0, \sigma)$
- [] $biodiversity_{ij} = \delta_j + \epsilon_i, \delta_j \sim N(0, \sigma)$

Question 8

Given the following data frame what would be the correct code to fit such model in R?

```
dat <- data.frame(plot = sample(1:100, 100),
                  department = paste0("D", sample(1:92, 100, replace = TRUE)),
                  biodiversity = rpois(100, lambda = 3))
head(dplyr::arrange(dat, plot))
```

```
  plot department biodiversity
1    1        D69            1
2    2        D76            5
3    3        D39            1
4    4        D71            5
5    5        D35            3
6    6        D15            1
```

- [] `lmer(biodiversity ~ 1 + (1 | department), data = dat)`
- [] `lmer(biodiversity ~ 1 + (1 | departement), data = dat)`
- [] `lmer(biodiversity ~ 1 + (0 + department), data = dat)`
- [] `lm(biodiversity ~ 1 + (1 | department), data = dat)`

Question 9

The model was fitted to the data and gives the following output:

```
Linear mixed model fit by REML ['lmerMod']
Formula: biodiversity ~ 1 + (1 | department)
   Data: dat

REML criterion at convergence: 386.5

Scaled residuals:
```

```
     Min       1Q   Median       3Q      Max
-1.85903 -0.64097 -0.07964  0.49833  2.84427

Random effects:
 Groups     Name        Variance Std.Dev.
 department (Intercept) 0.334    0.5779
 Residual               2.460    1.5686
Number of obs: 100, groups:  department, 64

Fixed effects:
            Estimate Std. Error t value
(Intercept)    3.142      0.176   17.85
```

What is the estimated standard deviation in average biodiversity between the department (rounded to two-digit)?

- [] 1.18
- [] 1.42
- [] 1.19
- [] 3.14

Question 10

We want to expand the analysis by estimating the effect of soil pH on forest biodiversity allowing the soil pH effect to vary between the department. Say that we have this dataset:

```
  plot department biodiversity        ph
1    9        D52            3  4.751179
2   79        D37            4  5.537854
3   17        D27            5  8.703592
4    7        D73            2  5.640032
5   94        D83            6 11.994288
6    8        D35            0 10.151230
```

Which is the correct code to fit a model with uncorrelated varying intercept and varying slope per department?

- [] `lmer(biodiversity ~ 1 + ph + (1 | department) + (0 + ph | department), data = dat)`
- [] `lmer(biodiversity ~ 1 + ph + (1 + ph | department), data = dat)`
- [] `lmer(biodiversity ~ 1 + ph + (1 | ph) + (0 + ph | department), data = dat)`

- [] `lmer(biodiversity ~ 1 + ph * department, data = dat)`